

Water Accounting (WA) Dashboard

A User Manual

September, 2026

1. What is a WA dashboard?

The water accounting dashboard provides detailed multi-year water accounts of a basin's, sub basin's and region's water resources including inflows, outflows, water use patterns and availability, to establish baseline conditions. It can also compare baseline and future water accounts (where available), to provide stakeholders with an understanding of the current and potential future water resources status.

The platform leverages advanced data visualization tools to offer user-friendly access to complex information, empowering stakeholders to understand the dynamics between various water-balance parameters. In addition, the dashboard was designed to ensure that it caters to stakeholders of diverse backgrounds and expertise, from policymakers to scientists. Stakeholder feedback was incorporated to enrich the platform's accessibility and user experience to enable active participation and engagement.

This web-based dashboard represents one of the tools that could be used in promoting effective stakeholder deliberation of water balance information for sustainable water resource management.

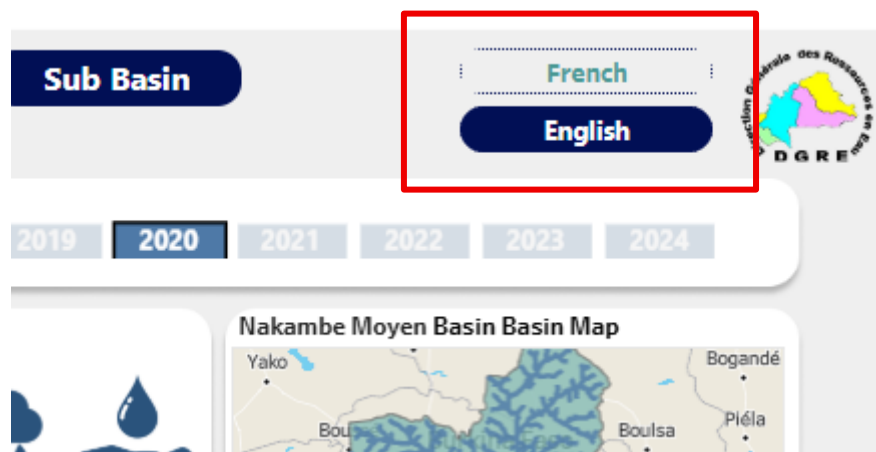
2. Why do we need a WA dashboard?

Water resource management is an essential global challenge, necessitating a comprehensive and inclusive approach to address the diverse needs of the stakeholders involved. Stakeholder deliberation is crucial in fostering cooperation and informed decision-making for sustainable water management. However, lack of hydrometeorological observations often limits our understanding on the complex interlinkages between competing demands for water in the basin or a country. To overcome the challenges of water data scarcity in managing water resources, the International Water Management Institute is generating reliable and systematic analysis ready water data products on water use, demand, water availability and scarcity using water accounting plus (WA+) framework. The water accounting plus (WA+) approach (Karimi et al., 2013) derives basin scale water availability and scarcity indicators using earth observation data products and limited in situ observations. The WA+ framework is an open-source python programming-based model that uses a collection of remote sensing data products and in situ data to quantify water accounting indicators such as a) water yield, b) irrigation/rainfed water use, c) productive/unproductive water use, and other water availability indicators at river basin scale. Continental water accounting plus (CWA+) is a modified version of WA+ where the water accounting indicators can be generated for any given boundary (catchment, or a country or a county).

The suite of water data generated by water accounting studies provides several new insights into water availability and scarcity indicators at continental scale. The availability of open source codes enable repeatable and rapid water accounting assessments without much effort. IWMI's WA dashboard is a web-based dashboard built using Tableau technology to enhance stakeholder engagement and facilitate effective deliberation by summarizing and communicating the key water accounts basin.

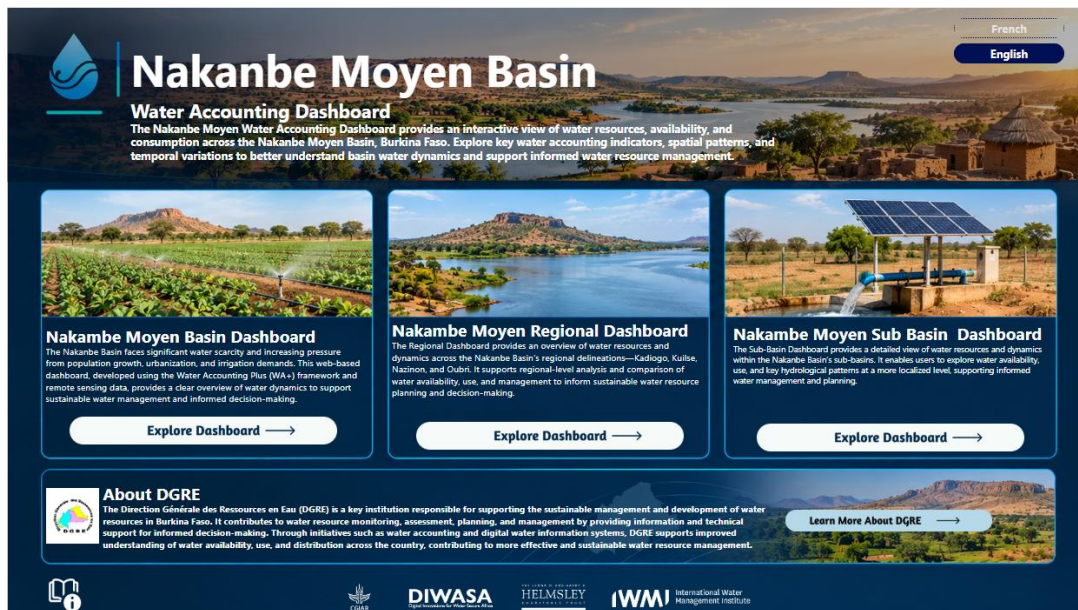
Switching Between Languages

English and French are the two languages in which the Nakanbe Dashboard is accessible. The dashboard's language option allows users to switch between English and French at any moment. Users can browse the dashboard and access the various water-related data and visualizations after choosing their favorite language.



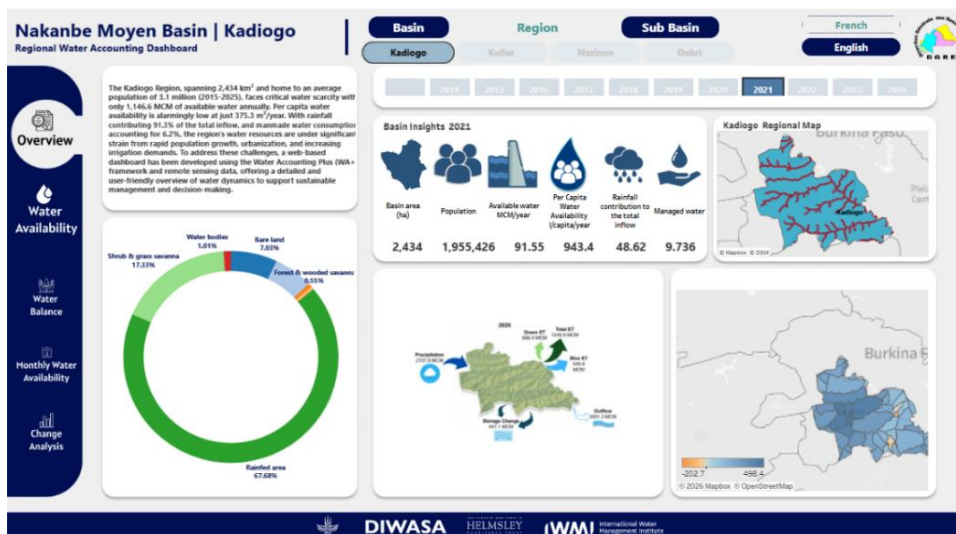
The dashboard is organized into **three main sections: Basin, Sub-basin, and Region**. Users can select any of these sections to explore the relevant information.





The landing page of the Nakanbe Moyen Basin Water Accounting Dashboard. At the top of the page, there is a large title along with language options (French and English) in the upper right corner. Below the title, a short introductory text explains that the dashboard provides interactive information on water resources, availability, and consumption in the Nakanbe Moyen Basin, Burkina Faso. The **Nakanbe Moyen Basin Dashboard** focuses on water dynamics across the entire basin by utilizing remote sensing and the Water Accounting Plus (WA+) framework to help address key challenges such as water scarcity and population pressure. For broader regional analysis, the **Nakanbe Moyen Regional Dashboard** provides an overview and comparison of water resources across specific areas, including Kadiogo, Kuilse, Nazinon, and Oubri. Finally, for detailed local management, the **Nakanbe Moyen Sub Basin Dashboard** offers a localized view of water availability, usage, and hydrological patterns within specific sub-basins. The Direction Générale des Ressources en Eau (DGRE) is a key government institution responsible for the sustainable management and development of water resources in Burkina Faso. It leads water resource monitoring, assessment, planning, and technical support to enable informed, data-driven decision-making.

2.1. Region Overview



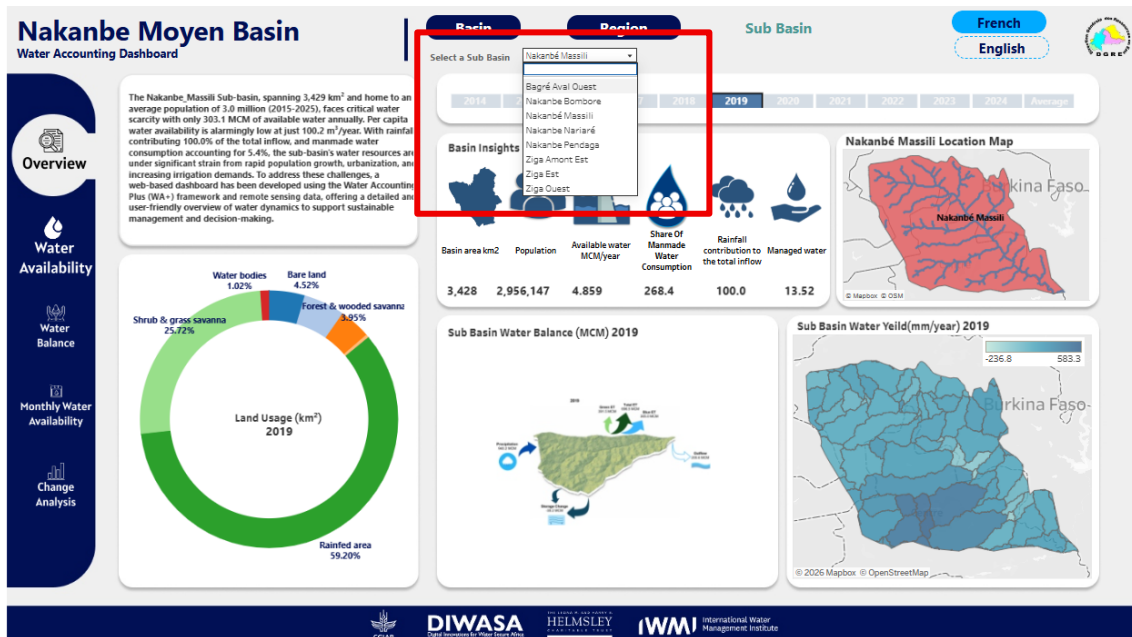
When the user clicks on the Region tab, the dashboard interface will be displayed with four available regions. The user can select any region to view the information related to that specific area.

When the user selects the first region, the dashboard will automatically update and display the data and indicators associated with that region.

Similarly, when the user selects another region, the dashboard interface will change accordingly and display the relevant information for the selected region. This feature allows users to easily explore

and compare water-related information across the four regions.

2.2. Sub Basin Overview



When the user clicks on the Sub-Basins tab, the Sub-Basins dashboard interface will be displayed. The dashboard provides information for eight different sub-basins, which can be selected using the drop-down menu.

The user can open the drop-down menu and select the required sub-basin from the available options. Once a sub-basin is selected, the dashboard will automatically update and display the relevant water-related information and indicators for that selected sub-basin.

This feature allows users to explore the water conditions, indicators, and other relevant information for each sub-basin individually. Users can also select different sub-basins from the drop-down menu to explore and compare the information available for each area.

3. Key Elements of the WA Dashboard

There are five key elements in a WA dashboard:

- Basin overview
- Water availability
- Water balance
- Monthly Water Availability
- Change Analysis
- Stationary index

3.1 Basin Overview

The landing page of the dashboard provides the basin overview information. Several key baseline statistics on the river basin are provided on this page. The description of each section is provided here.

The Nakanbe Basin, spanning N/A km² and home to an average population of 4.2 million (2014-2020), faces critical water scarcity with only 2.5 MCM of available water annually. Per capita water availability is alarmingly low at just 0.59 m³/year. With rainfall contributing 87.9% of the total inflow, and manmade water consumption accounting for 3.6%, the basin's water resources are under significant strain from rapid population growth, urbanization, and increasing irrigation demands. To address these challenges, a web-based dashboard has been developed using the Water Accounting Plus (WA+) framework and remote sensing data, offering a detailed and user-friendly overview of water dynamics to support sustainable management and decision-making.

Basin description: The basin overview page provides a brief synopsis of the dashboard. A basin description is provided to give a brief account of the basin hydrology and highlights important hydrologic challenges in the basin.

Figure 2. The description of the basin provided on the basin overview page

Year Selection Pane: Users can select a year from the pane. The dashboard will then update the information according to the selected year. Users can explore the dashboard and view the relevant data and visualizations for the selected year.

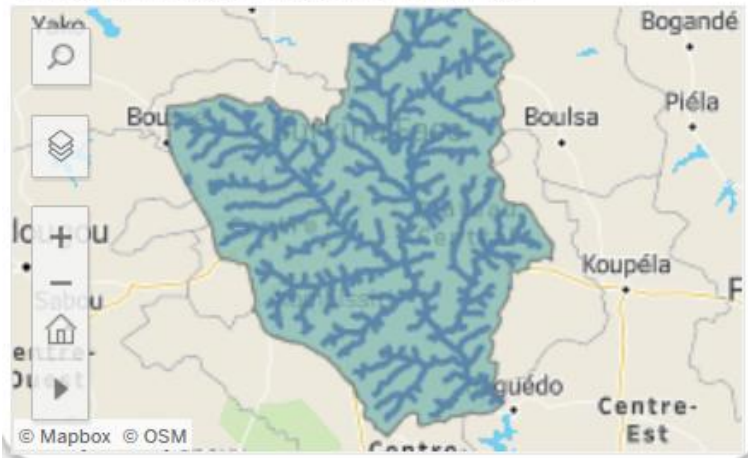


Basin Insights 2020

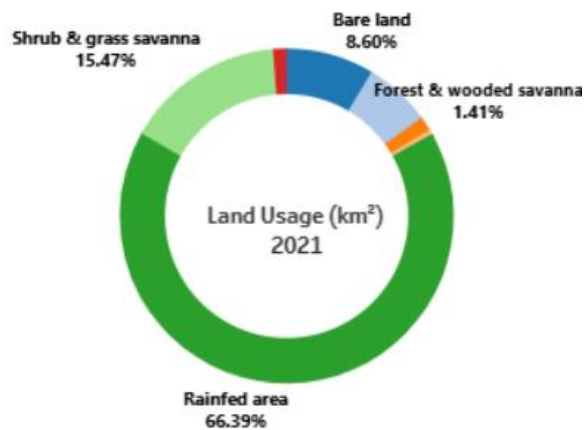
Basin area (km ²)	Population	Available water MCM/year	Per Capita Water Availability l/capita/year	Rainfall contribution to the total inflow	Managed water
12639.02	4,245,325	2.015	3.945	91.48	15.62

Basin Insights: The middle central portion of the basin overview page provides key information on basin insights such as basin area, population, available water, per-capita water availability, rainfall contribution to the inflow and managed water for future use. This Basin Insights section is interactively connected to the Year selection, and when a user clicks or changes the selected year, all the displayed data will dynamically update accordingly.

Nakambe Moyen Basin Basin Map



Basin Map: Here users can see an interactive map of the river basin. The users can zoom in and out of the basin area overlaid on the world map using the + and – symbols. The home icon on the map will reset the map to the full extent of the basin. The triangle icon offers additional features for interacting with the map.

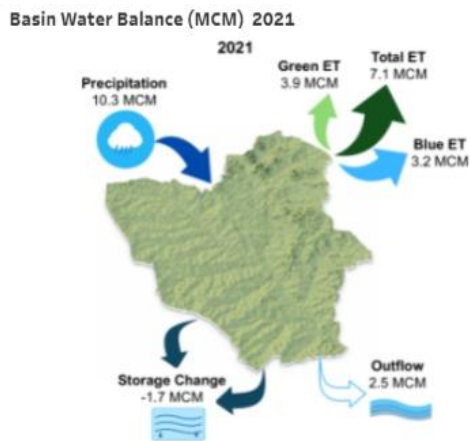


Land usage: The distribution of land use classes across the basin is presented using a double pie chart. In the outer pie chart, detailed land cover classes are reclassified into four broad Land Use for Water Accounting (LUWA) classes:

1. **Utilized Land:** Represents natural landscapes that are used in their natural form without modifying or altering water and land resources. For example, forests, grasslands, and shrublands utilized for grazing.
2. **Managed Water:** Represents areas where water is highly managed for agricultural activities, such as irrigated croplands.
3. **Modified Land:** Represents areas where land has been altered for human use. For example, natural landscapes cleared or modified to grow crops under rainfed conditions.
4. **Protected Land:** Represents areas designated as protected, such as national parks or other officially preserved zones.

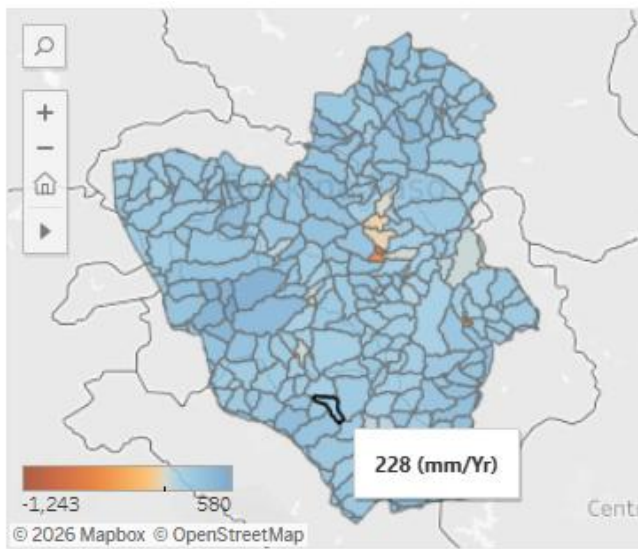
The inner pie chart shows the distribution of the detailed land cover classes that are grouped into these four broad LUWA categories.

This section is interactively connected to the Year selection, and when a user clicks or changes the selected year, all the displayed data will dynamically update accordingly.



Water Balance: The water balance figure shows modeled estimates of key water balance terms. Each figure is specific to the year selected on the year selection tab. The dark blue colored downward arrow represents the total volumetric annual precipitation (P, in km³) received in the basin. The light green colored upward arrow represents total volumetric evapotranspiration (TotalET, in km³). The lighter blue upward arrow represents BlueET (Blue ET, in km³), a portion of total ET, occurring from the blue water sources (surface water bodies, river, lakes or shallow groundwater aquifers). The small dark green

upward arrow represents, Rainfall ET or Green ET (in km³), a portion of total ET occurring from the green water sources (soil moisture replenished by the rainfall). The sum of Blue ET and Rainfall ET is equal to the total ET. The empty blue colored arrow represents basin outflow (in km³) and the storage change denotes the changes in the basin storage due to either groundwater abstraction (+ve value) or groundwater recharge (-ve value).

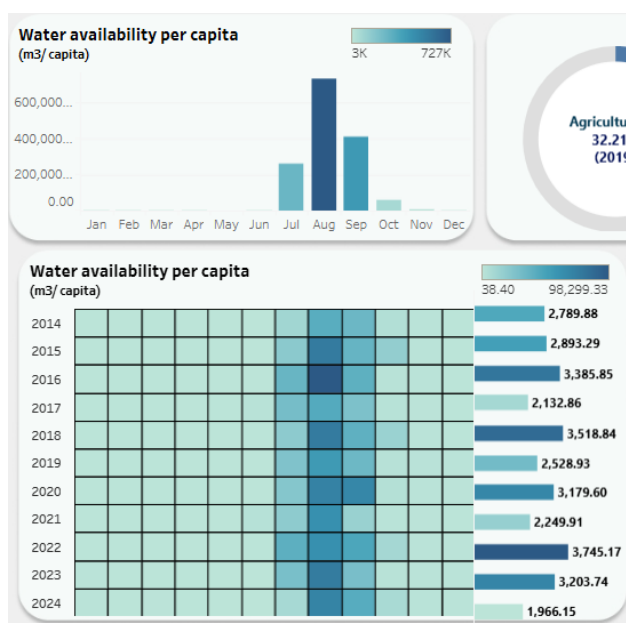


Water Yield: The map shows the water yield obtained from the water accounting analysis. The water yield is defined as water that is available after meeting landscape water requirement (landscape ET). This is the amount of water that can be exploitable for human needs. The map shows water yield for administrative regions within the basin. Regions show positive values (shades of blue) which indicate that at annual time scale, water availability for human needs is not a problem. Such

information is important to understand spatial variability of water availability within the basin. The blue areas are also called water towers of the basin are the regions that provide most water to the river and where future activities such as irrigation development or canals for diverting water can be constructed.

3.3. Water Availability

Information on water availability in a river basin is crucial for understanding various aspects of human, environmental, and economic well-being. Through this dashboard, water availability in a river basin is summarized using indicators of water availability for a) humans b) environment c) agriculture and d) other uses.



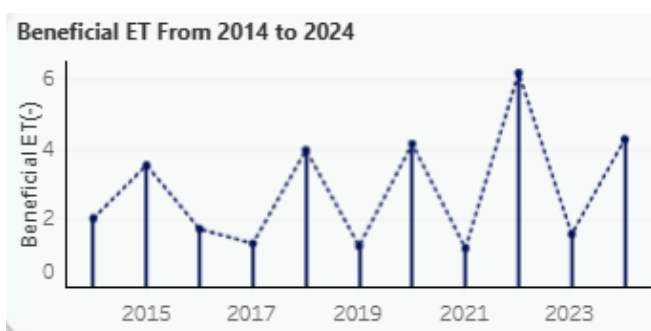
Water Availability Per Capita (m3/per capita): Water availability per capita refers to the amount of freshwater resources available for each person in a specific region or country. It is typically measured in cubic meters (m³) per person per year and is an important indicator of a region's or country's ability to meet the water needs of its population while also supporting economic and environmental demands. Calculating water availability per capita involves dividing the total annual freshwater resources of a region or country by its population. Within water accounting, we derive water availability per capita as

$$\text{Water Availability Per Capita} = \frac{\text{available water}}{\text{Population}}$$

The water availability per capita is estimated is presented at monthly and annual timescales from 2014 to 2024

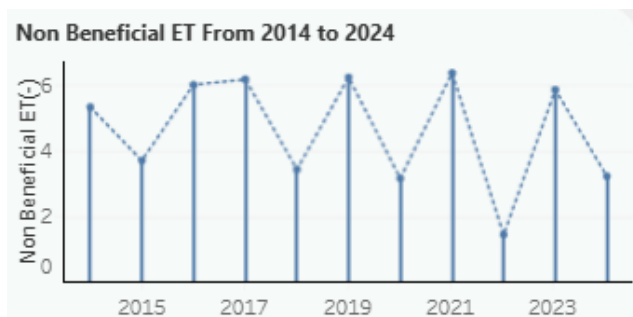
The annual total per capita water availability is presented to the right side as horizontal bar plots. The monthly average per capita water availability is presented on the top as the vertical bar plots.

Water Availability for further use (MCM)



Beneficial ET (BET)

Beneficial ET refers to the portion of evapotranspiration that contributes positively to ecosystem services or productive purposes. This includes ET from irrigated agriculture, rainfed crops, forests, and natural vegetation where water consumption supports biomass growth, food production, groundwater recharge, and ecological functions. Beneficial ET is considered “productive” water use.

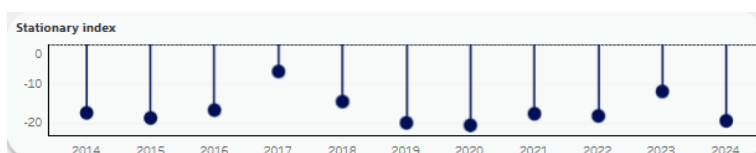


Non-Beneficial ET (NBET)

Non-Beneficial ET represents the portion of evapotranspiration that does not contribute to any productive use or ecological benefit. Examples include evaporation from bare soil, paved surfaces, open water bodies with no productive use, or excessive evaporation from poorly managed irrigation systems.

This water is essentially lost from the system without generating benefits.

Stationarity Index

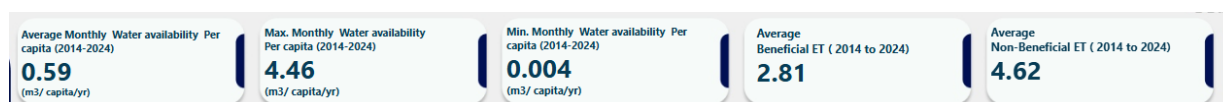


During the period from 2014 to 2024, the basin's stationarity index generally fluctuates between -6% and -20% from one year to the next, indicating a

constant trend toward the depletion of water resources: water withdrawals and losses (total evapotranspiration) exceed the recharge of groundwater and/or surface water reserves in any given year. Years in which the index drops to -20% indicate greater pressure on the resource than those at -6%, which could reflect year-to-year fluctuations in precipitation and/or high water demand. The recurrence of negative values from one year to the next warrants specific attention regarding the sustainability of current water use in the basin.

Stack Cards

This study card displays the studies available for the years **2014 to 2021**. It provides users with a quick overview of the studies conducted during these years.



Water Account Key Indicators



This section presents four circular charts that provide a quick overview of the selected year. The indicators include **Agricultural ET, Consumed Water, Managed ET, and Basin Closure**. Users can select

a year from the Year Selection drop-down menu, and the values in the circular charts will automatically update according to the selected year. These indicators help users quickly understand the main water account statistics and compare the performance of different indicators for the selected year. The circular charts provide a simple and visual way to explore the basin's water-related information.

Basin closure (%)

The basin closure is calculated as total amount of water available that is utilized within a basin.

$$\text{Basin closure} = \frac{\text{utilized}}{\text{available water}}$$

Any basin with estimates closer to 100% indicate basin closure – indicating most to all water availability in the basin is currently consumed within the basin. A smaller value indicates that water is available in the basin

3.2 Water Balance

The Water balance, also known as the hydrologic balance or water budget, is a fundamental concept in hydrology. It refers to the equilibrium or accounting of water inputs, outputs, and storage within a defined area, such as a watershed, catchment, or region. The water balance equation helps quantify the movement and distribution of water in various forms through the Earth's hydrological cycle.

Within the water accounting framework, water balance of a river basin is quantified and presented using a number of hydrologic variables. Unlike most hydrologic studies where the water balance is mostly represented by the key hydrologic variables such as Precipitation, evapotranspiration, discharge and change in storage, the water accounting framework derives a variety of hydrologic parameters. A full list of water balance indicators quantified in the water accounting framework are presented in Table 1.

The figure below shows basin input parameters on the left side and basin output parameters on the right side. The WA+ framework tracks both the flow and consumption (depletion) that occur within the basin as water moves from the inlet to the outlet of the basin. The depletion accounting is used to estimate how much of water is consumed over different landscapes.

This is summarized under four broad categories of land cover/land use – Protected-conservation areas with minimal changes in land and/or water management, Utilized- are areas with limited human influence and can include forest, natural pastures, savannahs and deserts, Modified- areas that are significantly modified by human activities usually for rainfed agriculture and Managed water use- are land use classes that are significantly modified for agriculture and include water purposefully withdrawn from the surface or groundwater sources for use. The flow accounting derives a bunch of parameters such as exploitable water, available water, managed water use, utilizable flows, non-utilizable outflows, reserved outflows, and non-consumed water.

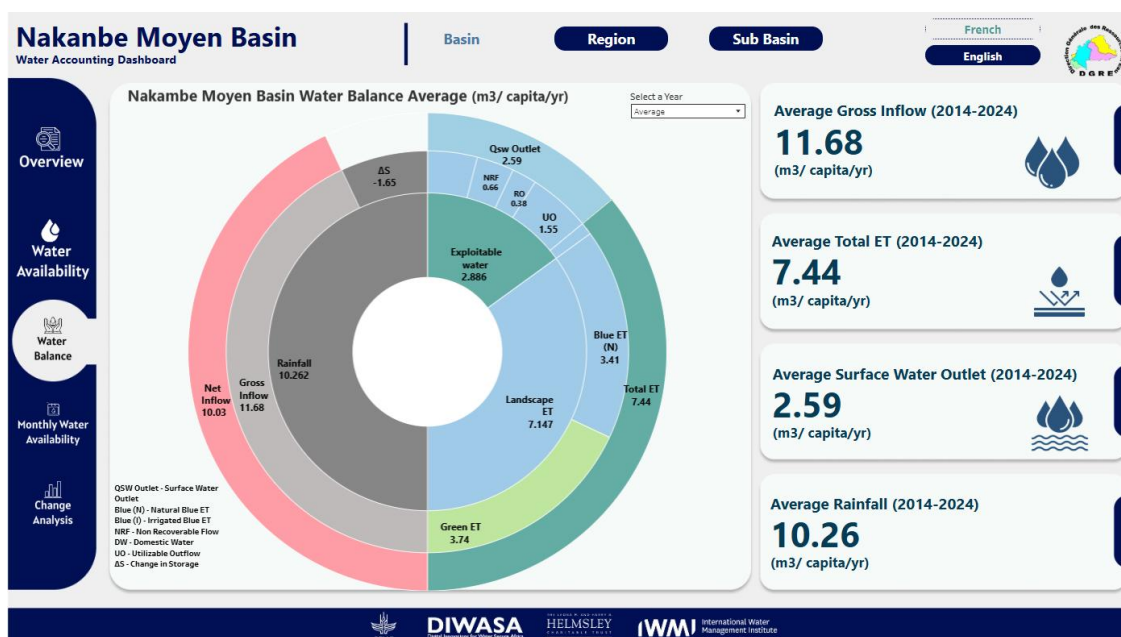
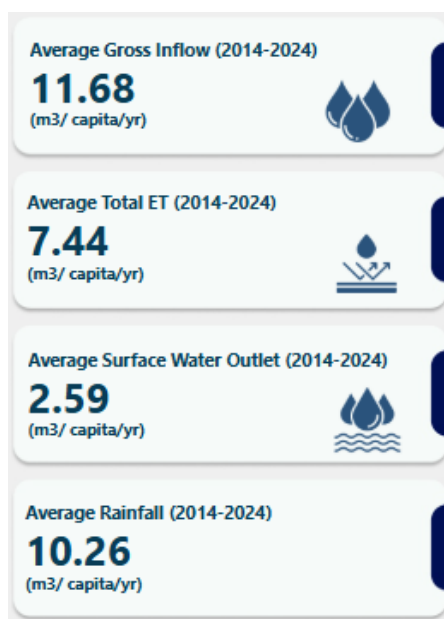


Table 1. List of hydrological variables and indicators quantified in the water accounting framework.

Stack Cards



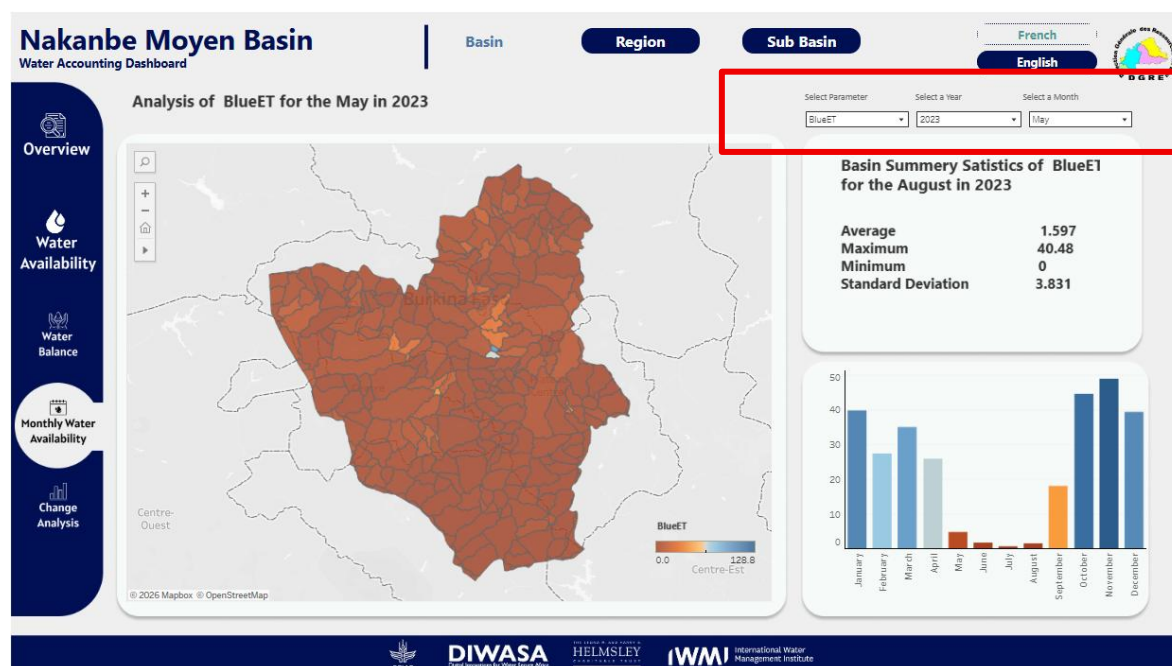
The four stack cards display key water-related indicators, allowing users to quickly understand the average values for the selected year or study period. The cards show the **Average Gross Inflow**, **Average Total Et**, **Average Water Outlet**, and **Average Rainfall** from 2014 to 2024.

These indicators provide a quick overview of the main water conditions and help users understand the water availability and movement within the selected area. By viewing these four cards together, users can easily compare the different indicators and gain a better understanding of the overall water situation.

No	Flux/Indicators	Description	Equation
1	P advection	Precipitation received in the basin, aggregated over the hydrologic year	$\sum_{i=1}^{12} P$
2	Basin inflow (interbasin transfer)	Surface water or groundwater diverted into the basin	Q_{in}^{sw} and Q_{in}^{gw} (Measured estimates)
3	Gross Inflow, GI	Total inflow from all sources	$P + Q_{in}^{sw} + Q_{in}^{gw}$
4	Change in the soil moisture, ΔSM	See equation 2.	See equation 2.
5	Net Inflow, NI	The gross inflow plus the change in soil moisture	$GI \pm \Delta SM$
6	ET rainfall, ET_{rain}	ETa that occurs from effective precipitation and canopy interception, summarized for all land cover classes (1 to n classes).	$\sum_{i=1}^n ET_{rain}$
7	ET incremental, ET_{incr}	ETa that occurs from other sources except effective precipitation and interception. Includes ET from irrigation water, groundwater abstraction, open water sources, summarized for all land cover classes (1 to n classes).	$\sum_{i=1}^n ET_{incr}$
8	Landscape ET, ET_{land}	ETa from natural landscapes (protected, utilized and modified land use classes); not due to water management.	$ET_{rain} + ET_{incr}$
9	Consumed water, C_{water}	Total ETa that occurs from all landscapes over all months	$\sum_{i=1}^{12} ET_a$
10	Utilized flow, $Uzed_{flow}$	ETa from managed water use (irrigated crops, managed reservoirs).	ET_{incr} from the managed water use class
11	Exploitable water, EX_{water}	The exploitable water is the amount of water that can potentially be used within the basin	$NI - ET_{landscape}$
12	Available water, AW	The water that is left after meeting ET and reserve flow requirements	$GI - ET_{land} - Reserve\ Flows$
13	Utilizable outflow, $Uzble_{flow}$	The water that can be reallocated for further uses after accounting for reserved flows and utilized flows.	$EX_{water} - ER_{flow} - Uzed_{flow}$
14	Qsw outlet	The river outflow at the outlet of the basin	Q_{outlet}^{sw}
15	Basin outflow (interbasin transfer)	Surface water or groundwater diverted to areas outside the basin	Q_{out}^{sw} and Q_{out}^{gw}
16	Non-consumed water, NC_{water}	Total outflow	$Q_{outlet}^{sw} + Q_{out}^{sw} + Q_{out}^{gw}$

3.3 Monthly water balance

The Monthly Change Analysis tab on the dashboard presents key water accounting variables such as Rainfall, Blue ET and Rainfall ET. A total of three parameters are available on the dashboard — Rainfall, Blue ET and Rainfall ET. All values are expressed in km³/year.



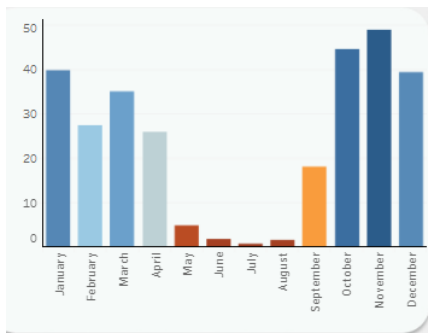
The Monthly Water Availability dashboard provides a visual overview of water availability within the selected region of the Nakanbe Moyen Basin. Users can select a parameter, year, and month from the drop-down menus to explore the corresponding water availability information. The main map displays the spatial distribution of the selected water availability parameter across the region, allowing users to identify variations between different areas.

Summery Stack Card



The Summary Stack Card presents the key statistical values for the selected parameter. It displays the average, maximum, minimum, and standard deviation, providing a quick summary of the selected data. These values help users understand the overall distribution, range, and variation of the selected parameter.

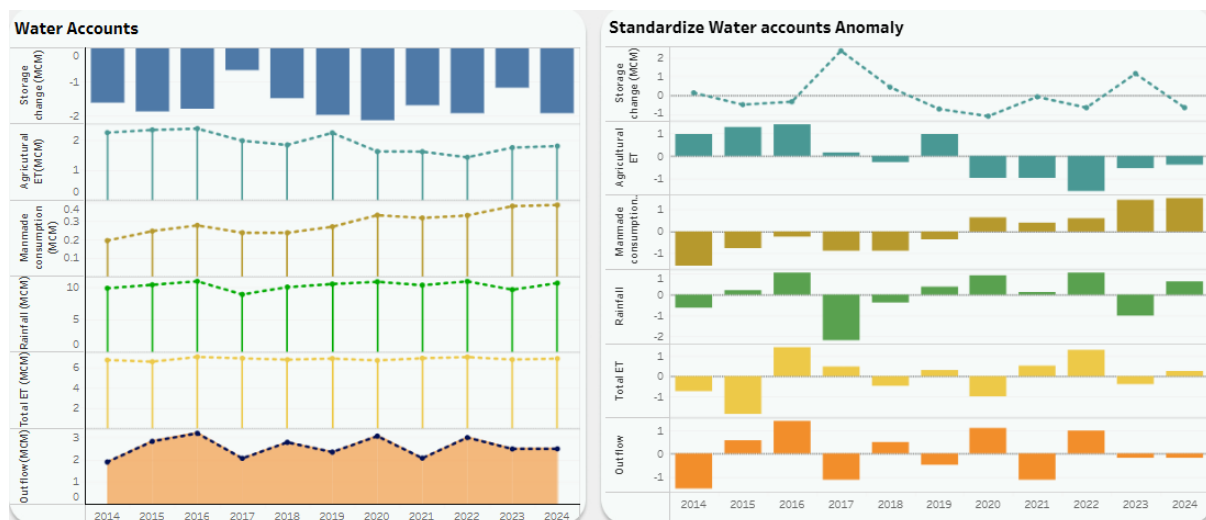
Monthly Parameter Histogram



The **monthly bar chart** represents the selected parameter across different months, allowing users to compare monthly values and identify variations over time. Users can use the **Parameter, Year, and Month** drop-down menus above the chart to select the required options. Based on these selections, the chart dynamically updates to visualize the corresponding data and helps users understand the monthly patterns and changes in the selected parameter.

3.4. Change Analysis

Long-term changes in hydrology refer to significant and persistent alteration in the water cycle and the distribution of water resources (both temporal and spatial) over extended periods of time. On the dashboard, we present charts. On the left side, we present long-term changes in hydrology with a focus on demonstrating increasing or declining trend in the parameter and on the right side, we present insights in to quantifying the change for 2014-2024.



The mean is calculated using all water account values, and the resulting mean value is displayed on the card.



Additional features on the dashboard:

There are several additional features available on the dashboard. A brief description and purpose of each of the icons located on the lower right corner are presented here.



Contact: The contact icon is located on the lower right corner of the dashboard and it would provide email information on whom to contact in case you have any questions on the dashboard.

About: The about icon provides more info on the project.

Printing options: The dashboard can be printed or saved using three options. The current view of the dashboard can be saved to the local computer in three different formats – PDF, JPG or PPT. Please use appropriate icon as per your need.

Download: Users can download displayed page data.

